

GLOBAL DISASTER SAFETY

Emergency Safety Guidelines

Comprehensive Instructions for 15 Natural & Environmental Disaster Types

International Emergency Preparedness Reference Document

This document provides comprehensive emergency safety guidelines for 15 disaster types as recognised by international disaster classification standards. Each chapter contains: overview and quick facts, before/during/after action guides, medical guidance, evacuation procedures, emergency contacts, and critical warnings.

15	8	7	WHO
Disaster Types	Action Phases	Sections Each	Aligned

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Reference Sources & Standards

This document draws on guidelines from the World Health Organization (WHO), International Federation of Red Cross and Red Crescent Societies (IFRC), United Nations Office for Disaster Risk Reduction (UNDRR), Federal Emergency Management Agency (FEMA), and national civil defence authorities. Statistical references reflect global disaster event records covering 1900-2021. This guide is intended for use by emergency preparedness professionals, community organisations, and individuals seeking authoritative disaster safety guidance.

Earthquake

Tectonic ground shaking — sudden, violent, no advance warning

Quick Facts & Classification

Onset	Sudden — seconds to minutes
Duration	Seconds to 2 minutes (main shock); hours to months (aftershocks)
Severity scale	Richter / Moment Magnitude (Mw); MMI for shaking intensity
Primary hazards	Ground shaking, surface rupture, liquefaction, landslides, tsunamis
Secondary hazards	Fire, dam failure, building collapse, infrastructure failure
Most affected	Urban areas on soft sediment; regions near tectonic plate boundaries
Global median deaths	20 per event (global); up to 87,000 in major events
Key terms	earthquake seismic tremor aftershock magnitude shaking structural collapse

Before: Preparedness Actions

These actions should be completed before a disaster event occurs. Preparedness is the most effective way to reduce injury, death, and economic loss.

- Conduct a structural assessment of your building for seismic vulnerability. Hire a licensed engineer to retrofit unreinforced masonry or weak soft-story structures.
- Secure heavy furniture (bookshelves, water heaters, refrigerators) to walls using anti-tip straps and L-brackets. Place heavy objects on lower shelves.
- Identify safe spots in every room: under sturdy tables, against interior walls away from windows. Practice Drop-Cover-Hold drills with all household members every 6 months.
- Prepare a Go-Bag containing: 72-hour water supply (4 litres per person per day), non-perishable food, first aid kit, copies of ID documents, cash, medications, torch, whistle, dust masks (N95), and a portable radio.
- Know how to shut off gas, water, and electricity at the main valves. Keep a wrench near the gas meter. Label all shut-off points clearly.
- Store emergency supplies in multiple locations (home, car, workplace). Do not rely on a single cache.
- Identify your local emergency assembly point and evacuation routes. Share the plan with all family members including children and elderly.
- Elevate water heaters and large appliances off the floor. Use flexible gas connectors approved for seismic zones.
- Keep shoes and a torch under your bed. Broken glass is the most common cause of foot injuries after an earthquake.

- Obtain earthquake insurance where available. Photograph and document all property for insurance claims.

During: Immediate Response Actions

These are the actions to take while the disaster event is actively occurring. Prioritise personal safety above all else.

- DROP immediately to hands and knees. This protects you from being knocked down and allows you to move if needed.
- Take COVER under a sturdy desk or table, or against an interior wall. Cover your head and neck with your arms.
- HOLD ON until the shaking stops. Move with the table if it shifts.
- If no table is available, crouch against an interior wall, covering your head and neck with both arms. Stay away from exterior walls, windows, and anything that could fall.
- If outdoors: move away from buildings, streetlights, and utility wires. Once in the open, drop to the ground and stay there.
- If in a vehicle: pull over away from buildings, bridges, overpasses, and power lines. Stay inside with seatbelt fastened until shaking stops.
- If in a coastal area: immediately after shaking, move inland and to higher ground — a tsunami may follow within minutes.
- Do NOT use elevators during or immediately after shaking.
- Do NOT run outside during shaking — most injuries occur when people try to move or run.
- Do NOT stand in doorways — this is an outdated myth. A doorframe offers no more protection than the rest of the structure.

After: Recovery Actions

These actions apply in the hours, days, and weeks following the disaster event. Recovery involves physical safety, health, and psychological wellbeing.

- Expect and prepare for aftershocks. They can occur minutes, days, or months after the main event and can be strong enough to collapse weakened structures.
- Check yourself and others for injuries. Provide first aid. Do not move seriously injured people unless they are in immediate danger.
- Check for gas leaks: smell for gas, listen for hissing. If suspected, open windows, leave immediately, and call the gas company from outside.
- Check for structural damage before re-entering your building. Look for cracks in walls, shifted foundations, and warped doors.
- Turn off electricity at the main breaker if you see sparks or damaged wiring. Do not use candles or open flames if gas leaks are possible.
- Do not use tap water until authorities confirm it is safe — water mains may be cracked.
- Tune in to official emergency broadcasts for damage reports, road closures, and shelter locations.
- Document all damage with photographs before cleaning up, for insurance and government assistance claims.

- Use text messages rather than voice calls to reduce network congestion. Social media check-ins can help families confirm safety.
- If displaced, go to official shelters. Do not enter red-tagged (unsafe) buildings under any circumstances.

Medical Guidance

- Crush syndrome: people trapped under rubble may develop life-threatening kidney failure when suddenly freed. Alert rescue teams — do not rapidly pull someone free without medical standby.
- Wound infection: debris-related wounds carry high infection risk. Clean all wounds thoroughly with clean water and apply antiseptic. Seek medical care for deep lacerations.
- Hypothermia: if trapped overnight, conserve body heat using any available materials — clothing, cardboard, insulation.
- Psychological trauma: acute stress reactions are normal. Provide calm reassurance. Children and elderly are especially vulnerable. Seek mental health support in the days and weeks following the event.
- Do not give food or water to an unconscious person. Place them in the recovery position if breathing.

Evacuation Procedures

- Evacuate only when authorities issue an official order or when your structure is clearly unsafe.
- Follow designated evacuation routes — do not use bridges or overpasses that may be structurally compromised.
- Drive slowly and watch for road damage, downed power lines, and sinkholes.
- If you smell gas anywhere along your route, do not use lighters or matches. Avoid the area and report it.
- Do not return home until authorities declare the area safe.

Emergency Contacts & Reporting

- National Emergency Management Agency (NEMA) — your country's primary earthquake emergency authority
- Local fire department — gas leaks, structural fires, trapped persons
- Red Cross / Red Crescent — shelter, family reunification, emergency relief
- Seismological monitoring centre — aftershock updates and magnitude reports

CRITICAL WARNING: NEVER re-enter a red-tagged building

A red tag means professional engineers have assessed the structure as unsafe. Re-entry risks fatal collapse during aftershocks. Respect all official tagging regardless of the perceived urgency of retrieving belongings.

SURVIVAL TIP: Whistle — your best survival tool

A loud whistle requires far less energy than shouting and can be heard by rescuers through rubble. Keep one in your Go-Bag, on your keyring, and under your bed. Three blasts is the universal distress signal.

Statistical Profile — Earthquake

Earthquakes account for 1,544 recorded events (1900–2021). Median deaths: 20 per event globally. Median affected: 4,050. Median damage: \$38.8M USD. Insurance coverage is only 17% — the lowest of all disaster types. Richter magnitude is the key impact predictor: Mw 7.0+ events have a 60x higher median death toll than Mw 5.0-6.0 events.

Flood

River overflow, flash floods, and coastal inundation — the world's most frequent disaster

Quick Facts & Classification

Onset	Slow-onset (river flood): hours to days; Flash flood: minutes to 1 hour
Duration	Hours to weeks depending on type and drainage capacity
Severity indicators	Flood depth, velocity, debris load, urban vs rural terrain
Primary hazards	Drowning, hypothermia, waterborne disease, structural collapse
Secondary hazards	Landslide, dam/levee failure, power outages, contaminated water supply
Most affected	Low-lying coastal and riverine communities; informal settlements without drainage
Global median deaths	16 per event (but mean is 1,735 due to extreme outliers)
Key terms	flood inundation flash flood river overflow waterlogging submerged drowning

Before: Preparedness Actions

These actions should be completed before a disaster event occurs. Preparedness is the most effective way to reduce injury, death, and economic loss.

- Know your flood zone. Check official flood hazard maps for your address. Flood zones change — re-check every 5 years or after major upstream development.
- Elevate electrical panels, HVAC systems, and appliances at least 30cm above the local base flood elevation. This single measure dramatically reduces flood damage.
- Install check valves in plumbing to prevent sewage from backing up into your home during floods.
- Keep important documents in a waterproof container or upload scanned copies to cloud storage. Include insurance policies, property deeds, passports, and medical records.
- Prepare a flood emergency kit: waterproof torch, life jackets for all household members, rope (15m minimum), first aid kit, water purification tablets, 3-day food and water supply, portable phone charger.
- Know multiple evacuation routes — floodwaters can cut off roads rapidly. Identify routes to high ground that do not cross low-lying bridges.
- Identify and pre-arrange an out-of-area contact person who can serve as the family communication hub.
- Keep sandbags available if you live in a frequently flooded area. Learn proper sandbagging technique — incorrectly placed sandbags are ineffective.
- Obtain flood insurance well in advance — standard property insurance does not cover flood damage in most countries.
- Trim trees and clear drainage ditches, gutters, and downspouts regularly to prevent blockage.

During: Immediate Response Actions

These are the actions to take while the disaster event is actively occurring. Prioritise personal safety above all else.

- Move immediately to higher ground when a flash flood warning is issued. Do not wait to observe the flood — by the time water is visible, escape routes may already be cut off.
- Never attempt to walk through moving water. Just 15cm of fast-moving water can knock an adult off their feet. 30cm of water can sweep away a vehicle.
- Never drive through flooded roads. 'Turn around, don't drown' — most flood deaths occur in vehicles. Water depth is impossible to judge on a road surface.
- If your vehicle is caught in rising water: exit immediately if water has not yet entered the cabin. If trapped, open a window and exit as soon as possible before water pressure makes doors impossible to open.
- Disconnect electrical appliances. Do not touch electrical equipment in wet conditions or if standing in water.
- If trapped on an upper floor or roof: signal rescuers with bright cloth, torch, or whistle. Do not attempt to swim unless in immediate danger from rising water.
- Avoid contact with floodwater — it may contain sewage, chemicals, and sharp debris.
- If outdoors and caught in a flash flood, climb to higher ground — seek a tree, hill, or elevated structure. Do not enter culverts, drainage ditches, or storm channels.

After: Recovery Actions

These actions apply in the hours, days, and weeks following the disaster event. Recovery involves physical safety, health, and psychological wellbeing.

- Do not return home until authorities declare it safe. Water may still be rising in some areas, and structural damage may not be visible.
- Assume all tap water is contaminated until officially cleared. Drink only bottled or boiled water. Use water purification tablets if available.
- Wear rubber boots and waterproof gloves when entering flooded buildings. Floodwater carries pathogens, chemicals, and sharp objects.
- Document all damage with photographs and video before cleaning. Notify your insurance company immediately.
- Discard all food that has come into contact with floodwater, including canned goods with dented or bulging lids.
- Dry out your home within 24-48 hours to prevent mould growth. Use fans and dehumidifiers. Mould can cause severe respiratory illness.
- Check for structural damage: cracks in foundations, warped floors, and shifted walls indicate unsafe structures.
- Watch for signs of waterborne disease in the following days: diarrhoea, vomiting, fever, skin rashes. Seek medical care promptly.
- Report downed power lines, broken gas mains, and road damage to authorities immediately.
- Complete an emotional recovery check with all family members — flooding is traumatic, especially for children.

Medical Guidance

- Drowning and near-drowning: all near-drowning victims require medical evaluation even if they appear recovered — delayed pulmonary oedema can be fatal.
- Wound infection: floodwater lacerations have a very high infection rate including tetanus. All wounds must be thoroughly cleaned and assessed by a medical professional.
- Waterborne diseases: leptospirosis, typhoid, hepatitis A, and cholera are common post-flood diseases. Symptoms appear 2-30 days after exposure. Seek medical care for any fever or gastrointestinal symptoms.
- Hypothermia: immersion in cold water reduces body core temperature rapidly. Remove wet clothing, wrap in dry blankets, and seek medical care.
- Carbon monoxide poisoning: portable generators used indoors or in garages after floods cause many post-flood deaths. Always run generators outdoors, at least 6m from any opening.

Evacuation Procedures

- Leave as soon as a flood WATCH is issued — do not wait for a WARNING. Early departure avoids traffic gridlock and road inundation.
- Take your emergency kit, medications, and important documents.
- Disconnect utilities at the main switches before leaving if time permits.
- Lock your home and follow the official evacuation route. Inform your out-of-area contact of your destination.
- If evacuating with pets, identify pet-friendly shelters or arrangements in advance — most emergency shelters do not accept animals.

Emergency Contacts & Reporting

- National Meteorological Service — official flood warnings and river level monitoring
- Local civil defence authority — evacuation orders, shelter locations
- Red Cross / Red Crescent — emergency shelter, food, and family tracing
- Local water authority — water safety announcements post-flood

CRITICAL WARNING: Flash floods kill in minutes

Flash floods are the deadliest form of flooding. The time from first warning to life-threatening flood depth can be under 10 minutes. Treat every flash flood warning as immediate. Never seek shelter in ground-floor rooms, underground car parks, or basements during a flash flood warning.

SURVIVAL TIP: The 30-cm rule for vehicles

30cm of moving floodwater is enough to float and sweep away a mid-size car. If the road ahead shows any standing water of uncertain depth, reverse and find an alternate route. No journey is worth a vehicle-related flood death.

Statistical Profile — Flood

Floods are the most frequent natural disaster globally: 5,551 events (1900–2021). Median deaths: 16 per event. Median affected: 11,000 — the highest of all disaster types. Median damage: \$49.5M USD. Flood events increased from 524 (1980s) to 1,725 (2000s) — a 3.3x increase over 20 years, driven by urbanisation and climate change.

DISASTER TYPE 03

Animal Accident

Dangerous encounters with wild or domestic animals — bites, attacks, and venomous stings

Quick Facts & Classification

Onset	Sudden — typically without warning
Types	Snake bite, scorpion sting, dog attack, large mammal attack, jellyfish sting, crocodile attack
Primary hazards	Venom, physical trauma, infection, anaphylaxis
Key regions	Sub-Saharan Africa, South Asia, Southeast Asia (highest snakebite incidence)
Global snakebite burden	~5.4 million bites per year; 81,000-138,000 deaths annually (WHO)
Key terms	snakebite animal attack venomous bite sting rabies anaphylaxis wildlife

Before: Preparedness Actions

These actions should be completed before a disaster event occurs. Preparedness is the most effective way to reduce injury, death, and economic loss.

- Learn to identify venomous species native to your region: snakes, spiders, scorpions, and marine animals. Local health departments and wildlife agencies provide identification guides.
- When working outdoors in high-risk areas: wear long trousers, closed-toe shoes or boots, and thick gloves. Use a torch at night — many venomous animals are nocturnal.
- Never put your hands in areas you cannot see: under rocks, into holes, into dense vegetation, or into dark spaces in buildings.
- Secure food and rubbish to avoid attracting wildlife near homes and camps.
- Know the location of the nearest hospital or clinic that stocks antivenom for locally prevalent species.
- Keep a first aid kit that includes clean bandages, antiseptic, and an antihistamine for mild allergic reactions.
- If you have a known allergy to insect stings (bees, wasps), carry a prescribed epinephrine auto-injector (EpiPen) at all times.
- Vaccinate domestic animals and yourself against rabies if living in or travelling to rabies-endemic regions.
- Keep children away from unfamiliar animals — even domestic ones. Teach children the 'stop, stay still, and slowly back away' rule when encountering an unfamiliar animal.
- Install door and window screens to prevent entry of scorpions and spiders into living spaces.

During: Immediate Response Actions

These are the actions to take while the disaster event is actively occurring. Prioritise personal safety above all else.

- **SNAKE BITE:** Stay calm. Movement accelerates venom spread. Immobilise the bitten limb at or below heart level. Remove constricting items (rings, watches, tight clothing) near the bite. Get to a medical facility immediately. Note the snake's appearance for identification — do not attempt to catch or kill it.
- **SCORPION STING:** Wash the sting site with soap and water. Apply a cold compress. Seek immediate medical care, especially for children and elderly — certain species produce systemic toxins requiring antivenom.
- **DOG ATTACK:** If approached by an aggressive dog — stand still, avoid direct eye contact, keep hands at your sides. If knocked down, roll into a ball protecting your face and neck.
- **LARGE MAMMAL (bear, elephant, hippo):** Follow species-specific guidance. For bears: do not run; back away slowly; use bear spray if available. For hippos or hippo warning signs: move away from the water edge immediately.
- **MARINE ANIMAL (jellyfish sting):** Remove tentacles carefully using tweezers or a card — not bare hands. Rinse with seawater (not fresh water, which can worsen stinging cell discharge). Seek medical care for widespread or severe stings.
- **ANAPHYLAXIS (severe allergic reaction to sting or bite):** Administer epinephrine auto-injector immediately. Call emergency services. Lay the person flat with legs elevated (unless breathing is compromised). Administer a second dose after 5-15 minutes if no improvement and EMS is delayed.

After: Recovery Actions

These actions apply in the hours, days, and weeks following the disaster event. Recovery involves physical safety, health, and psychological wellbeing.

- All animal bites and scratches require medical evaluation — even if the wound appears minor. Animal saliva carries bacteria causing dangerous infections including *Pasteurella*, *Capnocytophaga*, and tetanus.
- For snakebite: do NOT cut and suck the wound, apply a tourniquet, apply ice, or apply electric shock. These measures are ineffective or harmful. Keep the patient still and transport to hospital immediately.
- For any bite from a potentially rabies-carrying animal (dogs, bats, foxes, raccoons): begin post-exposure prophylaxis (PEP) within 24 hours if possible. Rabies is almost universally fatal once symptoms appear.
- Monitor the wound for signs of infection over 48-72 hours: increasing redness, warmth, swelling, pus, red streaking, or fever. Return to medical care immediately if these appear.
- Report all wild animal attacks to local wildlife authorities — particularly if the animal appeared sick (possible rabies) or if the attack was unprovoked.

Medical Guidance

- Antivenom is the only effective treatment for most serious envenomations. It must be administered in a clinical setting with resuscitation equipment available. Do not administer antivenom outside a hospital.
- Snakebite first aid: the pressure immobilisation bandaging technique is recommended for neurotoxic snakes (elapids) but contraindicated for cytotoxic snakes (vipers). Know which species in your region

require which technique.

- Anaphylaxis management: epinephrine is the first-line treatment — antihistamines alone are insufficient for anaphylaxis and must not substitute for epinephrine. Always call emergency services even after epinephrine administration.
- Wound management: all bite wounds should be irrigated thoroughly with clean water for at least 15 minutes to reduce bacterial load and rabies virus transmission.

Evacuation Procedures

- If in a wilderness area after a serious bite or sting, initiate evacuation to the nearest medical facility by the fastest available means.
- Immobilise the affected limb during transport. Keep the patient calm and still to slow venom circulation.
- If helicopter evacuation is needed, signal with bright materials or a mirror. Provide rescuers with the time of the incident and symptoms observed.

Emergency Contacts & Reporting

- Poison Control Centre — your country's 24-hour toxicology hotline for guidance on venomous bites and stings
- Nearest hospital with emergency antivenom stocks
- Local wildlife authority — to report attacks and receive guidance on dangerous local species

CRITICAL WARNING: Do NOT apply tourniquets or cut-and-suck for snakebites

Tourniquets cause tissue death and amputation. Cutting the wound introduces infection and does not remove venom. Both practices increase mortality. The only correct first aid for snakebite is immobilisation and rapid transport to a hospital with antivenom.

SURVIVAL TIP: Take a photo of the animal if safe to do so

Identifying the species is critical for selecting the correct antivenom. If you can safely photograph the animal from a distance without approaching it, do so. Do not attempt to capture or kill it — additional bites frequently occur during such attempts.

Statistical Profile — Animal Accident

Animal accidents include snakebite (highest burden), dog attacks, and large mammal encounters. WHO estimates 5.4 million snakebites annually, causing up to 138,000 deaths and 400,000 permanent disabilities. The global antivenom shortage is a critical healthcare challenge, particularly in sub-Saharan Africa and South Asia.

DISASTER TYPE 04

Drought

Prolonged water deficit — the silent, slow-onset disaster with the largest affected populations

Quick Facts & Classification

Onset	Slow — weeks to months to years
Duration	Months to years; some droughts persist for decades
Types	Meteorological (rainfall deficit), agricultural (soil moisture), hydrological (groundwater/river), socioeconomic
Primary hazards	Food insecurity, malnutrition, water scarcity, livestock loss, wildfire risk
Secondary hazards	Mass displacement, civil conflict, epidemic disease, economic collapse
Global median affected	758,000 per event — the highest of any disaster type
Global median deaths	143 per event (highest median of all disaster types)
Key terms	drought water scarcity food insecurity dry spell arid desertification famine malnutrition

Before: Preparedness Actions

These actions should be completed before a disaster event occurs. Preparedness is the most effective way to reduce injury, death, and economic loss.

- Monitor official drought indicators: Standardised Precipitation Index (SPI), Palmer Drought Severity Index, and seasonal rainfall forecasts from national meteorological services.
- Implement water conservation immediately when drought watches are issued: fix all leaks, reduce landscape irrigation, install water-efficient fixtures, and collect rainwater where legally permitted.
- Develop a household water budget. Know your daily per-person minimum requirement: 15 litres for basic survival, 50+ litres for health and dignity.
- Diversify food sources and maintain emergency food stocks covering at least 3 months. Prioritise calorie-dense, non-perishable items.
- For agricultural communities: shift to drought-resistant crop varieties, implement drip irrigation, and mulch soil to retain moisture.
- Know the location of alternative water sources: standpipes, water distribution points, NGO distribution schedules.
- Register vulnerable household members (elderly, infants, chronically ill) with local health authorities for priority access to drought relief programmes.
- Create or join a community water committee to coordinate equitable distribution of scarce water resources.

During: Immediate Response Actions

These are the actions to take while the disaster event is actively occurring. Prioritise personal safety above all else.

- Strictly ration all water use. The WHO minimum for drinking and sanitation is 15 litres per person per day. Prioritise drinking water, then hygiene, then food preparation.
- Use every drop: collect cooking water for cleaning, use grey water from washing for gardens (non-food plants).
- Prioritise vulnerable persons: infants, pregnant and breastfeeding women, the elderly, and those with chronic illness have the highest water and nutrition needs.
- Monitor children for signs of dehydration and malnutrition: sunken eyes, poor skin turgor, irritability, reduced urine output, weight loss. Seek medical care immediately.
- Protect remaining water sources from contamination. Cover wells, label all storage containers, and prevent animals from accessing water storage points.
- Avoid strenuous outdoor activity during peak heat hours (typically 10:00-16:00). Stay in shade or cool areas.
- For livestock keepers: destocking early is more economically advantageous than waiting for animals to die. Coordinate with agricultural extension officers on appropriate destocking strategies.
- Be alert to signs of social tension around water access. Conflict over scarce water is a documented drought-related risk.

After: Recovery Actions

These actions apply in the hours, days, and weeks following the disaster event. Recovery involves physical safety, health, and psychological wellbeing.

- When rainfall returns, do not immediately consume rainwater without testing or treating it — first rains after a prolonged drought can mobilise accumulated pollutants and pathogens from dry soil.
- Rehydration after severe dehydration should be gradual — oral rehydration solution (ORS) is safer than plain water for severely dehydrated individuals, especially children.
- Re-establish food gardens and agricultural production as soon as soil moisture permits, but prioritise fast-growing food crops over cash crops in the initial recovery period.
- Monitor for post-drought epidemic diseases: cholera, typhoid, and diarrhoeal diseases surge after droughts when compromised hygiene meets recovering water supply systems.
- Assess psychological impacts on all household members, especially children and those who lost livestock or crops. Drought-related economic loss is a major driver of depression and family breakdown.
- Apply for government and NGO drought relief and agricultural recovery assistance. Document all losses (livestock, crops, property) for claims.
- Invest recovery resources in long-term drought resilience: water harvesting systems, drought-resistant crops, and local water storage infrastructure.

Medical Guidance

- Severe dehydration in children: give Oral Rehydration Solution (ORS) — not plain water alone — in small, frequent sips. Seek medical care if the child cannot drink, has bloody diarrhoea, or deteriorates.

- Heat stroke: a core body temperature above 40°C with altered consciousness is a medical emergency. Cool the person immediately with any available method (water, shade, fanning) and seek emergency medical care.
- Malnutrition: mid-upper arm circumference (MUAC) is the key screening tool. A MUAC below 11.5cm in children 6-59 months indicates severe acute malnutrition requiring therapeutic feeding.
- Mental health: drought-related livelihoods loss is a leading driver of rural suicide in affected populations. Take all expressions of hopelessness seriously and connect people to mental health support.

Evacuation Procedures

- Evacuation during drought is primarily driven by food and water insecurity reaching critical levels.
- If water supply is completely exhausted, move to the nearest water distribution point or relief camp without delay.
- Organise community transport for those who cannot self-evacuate, particularly the elderly and disabled.
- Register at official displacement camps — this ensures access to food and water distribution, health services, and family tracing.

Emergency Contacts & Reporting

- National Meteorological Service — drought watch and rainfall forecast updates
- Ministry of Agriculture — destocking support, seed distribution, crop insurance
- UN World Food Programme (WFP) — food assistance in major drought events
- UNICEF — nutrition support for children and mothers

CRITICAL WARNING: Do not ignore early drought warning signals

Drought mortality is highest when communities and governments fail to act on early warning indicators. A 3-month delay in emergency response typically doubles the affected population requiring therapeutic feeding. Act on the first official drought watch — do not wait for declaration of emergency.

SURVIVAL TIP: ORS saves lives — prepare it correctly

Oral Rehydration Solution: dissolve 6 level teaspoons of sugar and 0.5 teaspoon of salt in 1 litre of clean water. This simple solution prevents death from dehydration. Prepare it fresh daily and give to those showing dehydration symptoms before reaching medical care.

Statistical Profile — Drought

Droughts affect more people than any other disaster type: median 758,000 per event. Global disaster records show 770 drought events (1900–2021). Deaths data is sparse (9% coverage) but median deaths of 143 reflects famine-related mortality. The 1983 Ethiopia drought caused ~3 million deaths, the largest single-event death toll in the dataset.

DISASTER TYPE 05

Epidemic

Rapid infectious disease spread — biological disasters requiring immediate containment

Quick Facts & Classification

Onset	Days to weeks depending on pathogen and transmission route
Duration	Weeks to years (pandemics)
Transmission routes	Airborne, droplet, contact, vector-borne, waterborne, foodborne, zoonotic
Key pathogens	Cholera, Ebola, influenza, COVID-19, dengue, yellow fever, meningitis, plague
Primary hazards	Illness and death; healthcare system overload; supply chain disruption
Secondary hazards	Economic collapse, food insecurity, social unrest, vaccine-preventable disease outbreaks
Classification	Biological disasters — viral, bacterial, parasitic, fungal, prion
Key terms	epidemic outbreak infection disease spread pathogen contagion quarantine isolation vaccination

Before: Preparedness Actions

These actions should be completed before a disaster event occurs. Preparedness is the most effective way to reduce injury, death, and economic loss.

- Maintain up-to-date vaccinations for yourself and all household members. Check the recommended vaccination schedule for your country and any travel destinations.
- Practise regular hand hygiene: wash hands with soap and water for at least 20 seconds, especially after using the toilet, before eating, and after contact with ill persons.
- Maintain a 30-day supply of prescription medications. Epidemic conditions can disrupt supply chains and pharmacy access.
- Stockpile basic medical supplies: thermometer, oral rehydration salts, paracetamol, antiseptic, gloves, and masks.
- Know the local epidemic warning system. Subscribe to alerts from your national health authority and WHO.
- Improve home ventilation: open windows and use fans to increase air circulation, reducing the risk of airborne pathogen accumulation.
- Safe food handling: refrigerate perishable foods promptly, cook meat to safe temperatures, separate raw from cooked food, and use safe water for food preparation.
- Manage standing water near your home to prevent mosquito breeding — mosquito-borne epidemics (dengue, malaria, Zika) are closely linked to vector habitat.
- Learn the signs of common epidemic diseases in your region so you can recognise them early and seek care promptly.

During: Immediate Response Actions

These are the actions to take while the disaster event is actively occurring. Prioritise personal safety above all else.

- Follow official public health guidance precisely and consistently. Measures such as isolation, quarantine, masking, and distancing are evidence-based and save lives.
- If you are ill: stay home. Do not go to work, school, or public gatherings. Notify your healthcare provider and follow their guidance for testing and treatment.
- If caring for an ill person at home: isolate them in a separate room if possible. Wear a mask. Wash hands thoroughly after any contact. Ventilate the room.
- Use personal protective equipment (PPE) correctly: masks must cover both nose and mouth; gloves should be removed safely to avoid self-contamination; eye protection is important for high-risk exposures.
- Maintain stringent hand hygiene during an epidemic. Avoid touching your face, particularly eyes, nose, and mouth.
- For vector-borne epidemics: use insect repellent (DEET or picaridin-based), sleep under insecticide-treated bed nets, wear long sleeves and trousers, and eliminate breeding sites.
- Maintain safe water and food practices even if the epidemic appears to be primarily transmitted by another route — compromised hygiene during epidemic conditions amplifies all disease risks.
- Accept vaccination when offered. Community-level vaccination (herd immunity) is the most powerful tool for epidemic control.

After: Recovery Actions

These actions apply in the hours, days, and weeks following the disaster event. Recovery involves physical safety, health, and psychological wellbeing.

- Continue all protective measures until officially declared by health authorities — many epidemic pathogens can linger for weeks after apparent resolution.
- Seek medical follow-up if you or a family member were ill, even if you recovered. Long-term sequelae are documented for many epidemic diseases.
- Participate in post-epidemic surveillance: report any new cases or unusual symptoms to health authorities. Rapid detection of a second wave is critical.
- Address mental health impacts: epidemic conditions are associated with high rates of anxiety, depression, and PTSD. Access available mental health services.
- Review and replenish your emergency medical stockpile. Assess what was lacking during the epidemic and adjust for future preparedness.
- Support community recovery: stigma against epidemic survivors (particularly Ebola, HIV, COVID-19) causes severe social harm. Actively counter misinformation and stigmatisation.

Medical Guidance

- Oral Rehydration Solution is the cornerstone treatment for most diarrhoeal epidemic diseases (cholera, rotavirus). Severe dehydration requires intravenous fluids in a clinical setting.

- Fever management: paracetamol (acetaminophen) is the safest first-line fever reducer. Avoid aspirin in children (Reye's syndrome risk) and ibuprofen in dengue (bleeding risk).
- Seek immediate medical care for: difficulty breathing, severe chest pain, confusion, inability to retain fluids, or high fever unresponsive to medication.
- Do not self-medicate with antibiotics. Antibiotic resistance is a global health crisis. Antibiotics are ineffective for viral diseases and must only be used on medical prescription.

Evacuation Procedures

- Evacuation during an epidemic must be carefully balanced against the risk of spreading the pathogen. Follow official public health guidance on movement restrictions.
- If evacuation is ordered: those who are ill should be segregated from healthy individuals during transport. Use masks and maintain distance.
- Report to official health screening points at evacuation centres — early case detection prevents spread within shelter populations.

Emergency Contacts & Reporting

- National health ministry or public health agency — official epidemic guidance and case reporting
- WHO — international health emergency guidance at [who.int](https://www.who.int)
- Poison control / Emergency hotline — for treatment guidance and reporting
- Local hospital — for testing, treatment, and referral

CRITICAL WARNING: Do not spread unverified health information

Misinformation during epidemics causes direct deaths by deterring people from seeking care, promoting dangerous home remedies, and undermining vaccination uptake. Share only information from verified health authorities. Report dangerous health misinformation when you encounter it.

SURVIVAL TIP: 20 seconds of handwashing prevents more disease than most medicines

Proper handwashing with soap and water for 20 seconds is the single most effective individual action to reduce epidemic disease transmission. Use alcohol-based hand sanitiser (at least 60% alcohol) when soap and water are unavailable. Teach children to sing a 20-second song while washing.

Statistical Profile — Epidemic

Epidemics (biological disasters) represent a significant and growing threat. COVID-19 (2020-2021) was the largest single epidemic event in modern history. Key intervention principles: early detection, rapid isolation, contact tracing, vaccination, and communication. The WHO International Health Regulations (IHR) provide the global framework for epidemic response.

DISASTER TYPE 06

Extreme Temperature

Heatwaves and severe cold events — silent killers that disproportionately affect the elderly

Quick Facts & Classification

Onset	Gradual over hours to days (but effects can become critical rapidly)
Types	Heatwave / heat dome; Cold wave / freeze; Wind chill
Duration	Days to weeks
Primary hazards	Heat stroke, heat exhaustion, hypothermia, frostbite
Most vulnerable	Elderly, infants, outdoor workers, athletes, chronically ill, those without air conditioning or heating
Global median deaths	38 per event; 2003 European heatwave killed ~70,000
Global median damage	\$200M USD — highest of all disaster types
Key terms	heatwave extreme heat cold wave hypothermia frostbite heat stroke heat exhaustion temperature

Before: Preparedness Actions

These actions should be completed before a disaster event occurs. Preparedness is the most effective way to reduce injury, death, and economic loss.

- HEAT: Identify the coolest locations in your community: air-conditioned public buildings, cooling centres, and shaded parks. Know their addresses and operating hours.
- HEAT: Install reflective window coverings or shade structures. Dark curtains absorb heat — use light-coloured, reflective materials.
- HEAT: Check on elderly neighbours and relatives daily during heatwave conditions. Isolation combined with extreme heat is fatal.
- COLD: Insulate your home before winter: seal draughts around doors and windows, insulate pipes in unheated spaces, and maintain your heating system annually.
- COLD: Prepare a cold emergency kit: blankets, warm clothing, hand warmers, non-perishable food, candles, matches, and a battery-powered radio.
- ALL: Know your local meteorological service's alert thresholds for extreme temperature warnings. Sign up for automated alerts.
- ALL: Review all medications — many drugs affect heat and cold tolerance: diuretics, beta-blockers, antipsychotics, antihistamines, and alcohol.
- ALL: Maintain adequate hydration throughout extreme temperature events. Thirst is a late indicator of dehydration.

During: Immediate Response Actions

These are the actions to take while the disaster event is actively occurring. Prioritise personal safety above all else.

- **HEAT:** Stay indoors in the coolest available space during the hottest hours (10:00-16:00). Avoid direct sun and physical exertion.
- **HEAT:** Drink water regularly — 2-3 litres per day minimum, more if active. Avoid alcohol, caffeine, and sugary drinks, which accelerate dehydration.
- **HEAT:** Wear loose, light-coloured, breathable clothing. Cover exposed skin and wear a wide-brimmed hat outdoors.
- **HEAT:** Use wet cloths, cool showers, and fans to lower body temperature. Mist spray bottles are highly effective.
- **HEAT:** Never leave children, elderly, or pets in parked vehicles — interior temperatures can reach 70°C within minutes even on mild days.
- **COLD:** Stay indoors. If you must go out, dress in dry, insulating layers (avoid cotton next to skin), cover all extremities, and limit exposure time.
- **COLD:** Eat calorie-rich food — your body burns more energy maintaining core temperature in cold. Stay hydrated — cold suppresses thirst sensation.
- **COLD:** If you detect shivering, confusion, slurred speech, or extreme fatigue in yourself or others — these are early hypothermia signs. Seek warmth immediately.

After: Recovery Actions

These actions apply in the hours, days, and weeks following the disaster event. Recovery involves physical safety, health, and psychological wellbeing.

- Check on vulnerable neighbours and community members as temperatures normalise.
- Seek medical evaluation for anyone who experienced severe heat or cold symptoms, even if they appear to have recovered.
- Replenish water and food supplies consumed during the event.
- Report deaths and severe illness associated with extreme temperatures to public health authorities — these events are frequently under-recorded.
- Assess your home for damage: frozen and burst pipes are a major post-cold-wave cause of property damage. Check your insurance coverage.

Medical Guidance

- **HEAT STROKE** (core temp above 40°C + altered consciousness): medical emergency. Cool immediately by any available method — immerse in cool water, wet clothing plus fanning. Call emergency services. This is life-threatening.
- **HEAT EXHAUSTION** (heavy sweating, weakness, cold and pale skin, weak pulse, nausea): move to cool environment, loosen clothing, apply cool wet cloths, give water. If symptoms progress to confusion or loss of consciousness, treat as heat stroke.
- **HYPOTHERMIA** (core temp below 35°C): move the person to a warm environment gently. Remove wet clothing. Cover with blankets. Give warm (not hot) liquids if conscious. Do not rub limbs vigorously. Call emergency services for severe hypothermia.

- FROSTBITE: do not rub frozen tissue. Rewarm only if there is no risk of refreezing. Rewarm in warm (37-40°C) water. Do not apply direct heat. Seek medical care — frostbite can result in tissue loss and amputation.

Evacuation Procedures

- If home heating fails during a severe cold event, evacuate to a warming centre, community centre, or hotel.
- If home cooling is inadequate during a severe heatwave, evacuate to an air-conditioned public building or cooling centre.
- Assist neighbours without transportation. Community members dying alone in their homes is the primary cause of extreme temperature mortality.

Emergency Contacts & Reporting

- National Meteorological Service — extreme temperature warnings
- Local emergency management — cooling and warming centre locations
- Emergency medical services — heat stroke and severe hypothermia are medical emergencies

CRITICAL WARNING: Elderly people may not feel extreme heat or cold accurately

Ageing impairs thermoregulation and thirst sensation. Elderly persons frequently do not report feeling too hot or too cold even when their core temperature is at dangerous levels. Check on elderly relatives and neighbours proactively — do not rely on them to self-report distress. Touch their skin and look for signs of confusion or extreme lethargy.

SURVIVAL TIP: The fan-plus-wet-skin technique is more effective than fan alone

A fan alone in extreme heat can accelerate fluid loss without significantly cooling. Mist exposed skin with water and then use a fan — evaporative cooling reduces skin temperature far more effectively than dry air movement. In extreme heat above 35°C, a fan alone may increase heat stress.

Statistical Profile — Extreme Temperature

Global disaster records show 603 extreme temperature events (1900–2021). Median deaths: 38 per event — one of the highest. Median damage: \$200,000 USD per event — the highest of all disaster types. The 2003 European heatwave (70,000 deaths) and 2010 Russian heatwave (56,000 deaths) represent the deadliest modern events. Deaths are massively underreported.

DISASTER TYPE 07

Fog

Dense atmospheric moisture reducing visibility — primarily a transportation and aviation hazard

Quick Facts & Classification

Onset	Rapid — can form in minutes, particularly radiation fog at night and dawn
Duration	Hours; typically dissipates by mid-morning
Types	Radiation fog, advection fog, upslope fog, sea fog (haar/fret)
Visibility threshold	Fog defined as visibility below 1km; dense fog below 200m; thick fog below 50m
Primary hazards	Road traffic accidents, maritime collisions, aviation incidents
Secondary hazards	Air quality deterioration (fog traps pollutants), cold-related illness in prolonged fog
Key terms	fog dense fog low visibility road accident maritime collision freezing fog smog

Before: Preparedness Actions

These actions should be completed before a disaster event occurs. Preparedness is the most effective way to reduce injury, death, and economic loss.

- Ensure all vehicle lights are fully functional: headlights, taillights, and fog lights (front and rear). Check and replace bulbs regularly.
- Maintain vehicle braking systems and tyres in good condition. Wet fog-coated roads are significantly more slippery than dry roads.
- Listen to official meteorological forecasts before long road or maritime journeys, particularly in seasons or locations prone to fog.
- If operating in maritime or aviation environments: ensure all navigation systems are functional and crew are briefed on fog procedures.
- Install reflective house numbers and gateposts — emergency vehicles need to locate properties rapidly even in zero-visibility conditions.
- Know the fog warning thresholds in your region. Many countries issue fog warnings at 200m visibility.

During: Immediate Response Actions

These are the actions to take while the disaster event is actively occurring. Prioritise personal safety above all else.

- ROAD: Reduce speed significantly — visibility in dense fog can be reduced to under 50 metres, leaving insufficient stopping distance at normal speeds.

- ROAD: Use low-beam headlights and rear fog lights (where fitted). High beams reflect off fog and reduce visibility further.
- ROAD: Increase following distance to at least 3 times the normal distance. Maintain awareness of vehicles behind you.
- ROAD: Use road markings and edge lines as navigation guides. In extreme conditions, pull off the road completely, switch off your lights (to avoid being hit by drivers following your lights), and activate hazard lights.
- ROAD: Do not overtake in fog. Do not stop on the carriageway unless absolutely necessary.
- MARITIME: Reduce speed to safe manoeuvring speed. Sound fog signals as required by collision regulations (COLREGs). Maintain a full radar watch and post additional lookouts.
- AVIATION: Follow instrument flight rules (IFR) procedures. Do not attempt visual approaches below minima. Comply with ATC instructions at all times.
- PEDESTRIAN: Wear high-visibility clothing. Use pavements and cycle paths. Make yourself as visible as possible to drivers.
- If you are caught in sudden dense fog while walking in an unfamiliar area, stop and wait for conditions to improve rather than continuing and becoming disoriented.

After: Recovery Actions

These actions apply in the hours, days, and weeks following the disaster event. Recovery involves physical safety, health, and psychological wellbeing.

- Allow extra time for road surfaces to dry after dense fog — residual moisture can create slippery conditions for hours after fog lifts.
- Check for road accident reports on your intended route before driving in the post-fog period — accident scenes create significant secondary hazards.
- If you experienced a near-miss or accident in fog, report it to the relevant transport authority. Fog-related accident data helps authorities identify high-risk locations for infrastructure improvements.

Medical Guidance

- Freezing fog (when temperatures are at or below 0°C): exposed skin and surfaces become coated in ice. Treat any ice coating on vehicle windscreens and mirrors before driving. Treat personal exposure as a cold weather risk.
- Air quality impacts: fog can trap urban pollutants, creating smog conditions. Persons with respiratory conditions (asthma, COPD) should remain indoors during dense smog-fog events and have rescue medication accessible.

Evacuation Procedures

- Fog rarely requires evacuation. If evacuation is required during fog conditions (due to a concurrent emergency), drive only at safe speeds appropriate to visibility.
- Emergency services may be delayed in dense fog. Call emergency services immediately for any life-threatening situation but understand that response times may be extended.

Emergency Contacts & Reporting

- National Meteorological Service — fog warnings and visibility forecasts
- Road traffic authority — accident reporting and road condition updates
- Maritime rescue coordination centre — for fog-related maritime emergencies

CRITICAL WARNING: Pile-up risk in fog: the chain reaction crash

Multi-vehicle motorway pile-ups in fog are almost always caused by drivers maintaining normal speeds and following distances. A single incident becomes a chain crash as subsequent vehicles, unable to stop in time, rear-end the queue. If you are forced to stop on a motorway in fog, exit your vehicle and move to the hard shoulder or safety barrier — do not remain inside the vehicle.

SURVIVAL TIP: Slow down more than you think necessary

Human perception consistently underestimates driving speed in fog. At 80 km/h in fog with 100m visibility, stopping distance is 55m — leaving no margin. The correct response is to reduce speed until you can comfortably stop within your visible distance. Most drivers do not slow enough.

Statistical Profile — Fog

Fog-related disasters are primarily mass transport accidents. High-profile events include the 1952 London Great Smog (4,000 direct deaths, 8,000 delayed deaths) and major motorway pile-ups across Europe, North America and Asia. Fog is climatically increasing in certain regions as urban heat islands alter local meteorology.

DISASTER TYPE 08

Glacial Lake Outburst Flood

Sudden catastrophic release of glacially dammed water — GLOF events in mountain regions

Quick Facts & Classification

Onset	Rapid to sudden — can progress from warning signs to catastrophic flood in hours or days
Primary mechanism	Moraine dam failure, ice dam failure, or subglacial drainage; triggered by slope failure, ice avalanche, or seismic activity
Regions at risk	Himalayas, Karakoram, Hindu Kush, Andes, Alps, Central Asia
Primary hazards	Extreme flash flooding, debris flows, massive sediment deposition
Secondary hazards	Landslides, bridge and road destruction, prolonged river disruption
Classification	Hydrological disaster — flood subtype
Key terms	glacial lake outburst GLOF moraine dam glacier flood mountain flood ice dam

Before: Preparedness Actions

These actions should be completed before a disaster event occurs. Preparedness is the most effective way to reduce injury, death, and economic loss.

- If you live downstream of a glacial lake system, identify whether your community is in a GLOF hazard zone. National geological and water resource agencies typically maintain GLOF hazard maps.
- Know the early warning signs of potential GLOF: unusual sounds from upstream (rumbling, cracking), sudden increase or decrease in river flow, and unusual turbidity (cloudiness) of river water.
- Know your evacuation route to high ground. In mountain valleys, the evacuation window may be extremely short — practise your route.
- Register with local emergency management for early warning notifications. Many GLOF-prone regions have installed automated early warning systems.
- Avoid camping, farming, or building on valley floors or near river banks in GLOF-prone areas, particularly during the melt season (typically April-September in the Northern Hemisphere).
- Keep a Go-Bag ready during the melt season if you live in a GLOF risk zone.

During: Immediate Response Actions

These are the actions to take while the disaster event is actively occurring. Prioritise personal safety above all else.

- If a GLOF warning is issued or you observe GLOF warning signs: evacuate immediately to high ground without waiting for official confirmation.

- Move perpendicular to the valley and uphill — the flood surge travels at high speed along the valley floor.
- Do not attempt to cross rivers or streams during or after a GLOF — conditions are rapidly changing and debris loads are extremely dangerous.
- The flood surge in a GLOF can carry enormous boulders, logs, and debris at high velocity. Do not underestimate the force.
- If caught in a GLOF flood without time to evacuate to high ground: seek the highest available position immediately — trees, buildings, or elevated terrain features.
- Alert others downstream by any available means: phone, radio, horn, or shouting.

After: Recovery Actions

These actions apply in the hours, days, and weeks following the disaster event. Recovery involves physical safety, health, and psychological wellbeing.

- Do not return to valley floor until authorities confirm the GLOF has subsided and the area is safe.
- River channels may be fundamentally altered by sediment deposition. Old bridges, ford crossings, and paths may no longer be safe.
- Report any structural damage to bridges, roads, and irrigation systems to authorities immediately.
- Monitor river flows for days to weeks after a GLOF — repeated surge pulses can occur as additional dam material fails.
- GLOFs can destroy water supply infrastructure. Assume all local water sources are contaminated until tested.

Medical Guidance

- GLOF flood victims face similar medical challenges to flash flood victims: drowning, hypothermia, and severe traumatic injury from debris impact.
- Cold water immersion is especially dangerous in mountain glacial floodwater. Hypothermia can develop in minutes. Remove wet clothing, insulate, and seek medical care.
- Sediment-contaminated water carries pathogens. All wound care must be thorough and all water must be treated before consumption.

Evacuation Procedures

- Immediate high-ground evacuation is the only effective response to an imminent GLOF. There is no preparation that can protect structures in the valley floor from a major GLOF.
- Evacuation routes in mountain terrain should be pre-established and practiced. Improvised evacuation in steep terrain is extremely dangerous.
- Community early warning and evacuation systems (including manual sirens and community alarm trees) have dramatically reduced GLOF mortality in well-prepared communities in Nepal and Pakistan.

Emergency Contacts & Reporting

- National glaciology and water resources department — GLOF monitoring and early warning

- Local emergency management — evacuation orders
- ICIMOD (International Centre for Integrated Mountain Development) — regional GLOF expertise and early warning systems

CRITICAL WARNING: GLOFs can travel at 15-30 metres per second

A glacial lake outburst flood surge travels at the speed of a highway vehicle. The time between audible warning (deep rumbling sound) and arrival of the flood wall can be less than 5 minutes in confined valleys. There is no time to collect belongings. Evacuation to high ground must be immediate and instinctive.

SURVIVAL TIP: Unusual river behaviour is your earliest warning

GLOFs are often preceded by hours of unusual river behaviour: sudden flow increase or decrease, unusual turbidity, and floating ice debris. If river behaviour seems abnormal during the melt season, inform authorities and move to high ground immediately as a precaution.

Statistical Profile — Glacial Lake Outburst Flood

GLOFs are an increasing concern as climate change accelerates glacial melt. The Himalayan region alone contains over 15,000 glacial lakes, of which approximately 200 are considered potentially dangerous. Major historical GLOFs include the 1985 Dig Tsho GLOF in Nepal and multiple events in Pakistan's Gilgit-Baltistan region.

Impact Event

Extraterrestrial object impacts — meteorites, bolides, and near-Earth object threats

Quick Facts & Classification

Types	Meteorite fall, bolide (fireball), airburst, small asteroid impact
Onset	Sudden and virtually without practical warning for small events
Frequency	Meteorite falls: multiple per year globally; significant airburst events: every 10-100 years; Tunguska-scale events: every 500-1000 years
Historical events	Tunguska 1908 (12 megatons, 2,000 km2 forest destroyed); Chelyabinsk 2013 (1,500 injured by shockwave)
Primary hazards	Shockwave/airburst overpressure, thermal radiation, debris impact, secondary fires
Key terms	meteorite impact bolide fireball airburst shockwave extraterrestrial impact space rock

Before: Preparedness Actions

These actions should be completed before a disaster event occurs. Preparedness is the most effective way to reduce injury, death, and economic loss.

- For meteorite impacts, practical individual preparedness is limited — space agencies (NASA, ESA) provide the primary monitoring and early warning function through planetary defence programmes.
- For minor meteorite fall events: secure structures, vehicles, and outdoor equipment during periods of known meteor shower activity.
- For airburst events (with very short warning): the primary protective action is to move away from windows immediately.
- Follow space agency announcements for any near-Earth object (NEO) that receives media coverage. Distinguish between media speculation and official risk assessments.
- General household emergency preparedness (Go-Bag, emergency supplies) is applicable to the aftermath of any significant impact event.

During: Immediate Response Actions

These are the actions to take while the disaster event is actively occurring. Prioritise personal safety above all else.

- If you observe a large fireball or hear a detonation sound: move immediately away from windows and glass surfaces. The Chelyabinsk airburst injured 1,500 people, mostly from broken glass caused by the shockwave arriving approximately 2 minutes after the visual event.
- Drop to the floor and cover your head and neck. Shield yourself from broken glass.

- If outdoors: lie flat on the ground, away from buildings and glass, facing away from the direction of the fireball.
- After the initial shockwave, be alert for secondary waves — large airburst events can produce multiple shockwave pulses.
- Stay away from the impact site or crater area until emergency services arrive and assess radiation, chemical, or biological contamination risks from the object.

After: Recovery Actions

These actions apply in the hours, days, and weeks following the disaster event. Recovery involves physical safety, health, and psychological wellbeing.

- Treat the aftermath as a disaster scene. Fires, structural collapse, and infrastructure damage are the primary post-impact hazards.
- Do not approach or handle meteorite fragments — they may be hot and may have sharp edges. Report their location to authorities.
- If impact caused fires: follow wildfire or urban fire evacuation procedures.
- Monitor official broadcasts for hazard assessment and guidance from emergency management authorities.

Medical Guidance

- Blast overpressure injuries: concussion, ruptured eardrums, pulmonary damage. Protect ears and seek medical evaluation after any significant shockwave exposure.
- Glass injury: airburst events primarily injure through shattered window glass. Deep lacerations require immediate medical attention.
- Flash blindness: a large bolide can temporarily blind observers. Do not look directly at an exceptionally bright fireball.

Evacuation Procedures

- For significant impact events, follow official emergency management evacuation orders.
- Impact sites may have hazardous contamination — maintain a wide exclusion zone until officially cleared by emergency authorities.

Emergency Contacts & Reporting

- National emergency management agency — post-impact response coordination
- NASA Jet Propulsion Laboratory Center for Near Earth Object Studies (CNEOS) — official near-Earth object monitoring
- Local emergency services — immediate post-impact response

CRITICAL WARNING: Do not rush to the impact site

Impact sites attract large numbers of bystanders who place themselves at risk from secondary hazards including debris, fire, unstable ground, and potential contamination. Emergency services require unrestricted access. Stay away from impact sites until officially declared safe.

SURVIVAL TIP: Move away from windows immediately upon seeing a large fireball

The Chelyabinsk 2013 airburst demonstrated that the shockwave arrives 1-3 minutes after the visual event. This window allows time to move away from windows if you act immediately upon seeing an unusually bright fireball. Do not stop to film it — move first.

Statistical Profile — Impact Event

Impact events are rare in Global disaster records show but historically significant. Most recorded events are meteorite falls without casualties. The risk of a civilisation-threatening impact is real but extremely low on human timescales. Planetary defence through early detection (NASA's DART mission) and deflection is the primary risk mitigation strategy.

DISASTER TYPE 10

Insect Infestation

Locust swarms and pest outbreaks — agricultural disasters threatening food security

Quick Facts & Classification

Primary pest	Desert locust (<i>Schistocerca gregaria</i>) — most destructive migratory pest globally
Swarm size	A 1 km ² swarm contains ~80 million locusts consuming as much food per day as 35,000 people
Regions at risk	Sub-Saharan Africa, Middle East, South Asia, North Africa — the 'locust belt'
Onset	Can be rapid — swarm arrival warning may be hours to days
Duration	Days to months for an infestation; locust upsurges persist for years
Primary hazards	Complete crop destruction, pasture loss, food insecurity, livestock starvation
Key terms	locust swarm insect infestation pest outbreak crop destruction food security plague

Before: Preparedness Actions

These actions should be completed before a disaster event occurs. Preparedness is the most effective way to reduce injury, death, and economic loss.

- Monitor FAO Desert Locust Information Service (DLIS) bulletins and national agricultural authority warnings in locust-prone regions.
- Maintain emergency food stores covering at least 3 months of household consumption in high-risk regions.
- Coordinate with community agricultural extension officers on approved pesticide stockpiles and aerial spraying request procedures.
- Protect stored grain against insect infestation using approved hermetic storage methods (grain bags, metal silos). Hermetically stored grain is inaccessible to most insects.
- Identify alternative income sources and social safety net programmes that can be activated during a locust-driven food crisis.
- Document baseline crop condition with photographs for insurance and disaster relief claims.

During: Immediate Response Actions

These are the actions to take while the disaster event is actively occurring. Prioritise personal safety above all else.

- Alert national agricultural authorities immediately upon observing locust swarms — early reporting enables coordinated aerial response.
- Cover stored food, animal feed, and water sources to prevent contamination by insect bodies and frass.

- Protect vegetable gardens with physical barriers (nets, row covers) — large-scale field crops cannot be protected by physical means.
- Livestock: move animals away from infested fields immediately. Excessive ingestion of locusts can cause digestive problems in some animals.
- Do not use unauthorised or household pesticides on large swarms — they are ineffective at swarm scale and hazardous to people and livestock.
- Cooperate fully with authorised aerial or ground control operations — compliance and clear access are essential for effective pesticide application.
- Document all crop losses with dated photographs for disaster relief and insurance claims.

After: Recovery Actions

These actions apply in the hours, days, and weeks following the disaster event. Recovery involves physical safety, health, and psychological wellbeing.

- Apply for agricultural disaster relief from government and international agencies (FAO, WFP). Timely application is critical — relief programmes have limited windows.
- Assess and document all crop and livestock losses thoroughly for claims and assistance programmes.
- Plant fast-maturing crop varieties to reduce the period of food insecurity following the infestation.
- Clean up locust carcasses from water sources before they decompose and cause water contamination.
- Participate in community replanting and seed distribution programmes. Coordinated community replanting maximises the efficiency of seed and input support.
- Monitor for secondary pest outbreaks — disrupted agricultural ecosystems after locust events are vulnerable to follow-on pest explosions.

Medical Guidance

- Locusts are safe to eat and are a nutritious, protein-rich food source consumed across Africa, the Middle East, and Asia. Collection and consumption of locusts during an infestation can be a legitimate food security strategy, provided the insects have not been treated with pesticides.
- Pesticide exposure: aerial spraying operations use organophosphate and pyrethroid pesticides. Avoid sprayed areas for the period specified by the spray team. Wash all produce from sprayed areas thoroughly before consumption.
- Allergic reactions to locusts are rare but possible in persons with pre-existing seafood allergies (locusts share allergens with crustaceans).

Evacuation Procedures

- Evacuation is rarely required during locust events. However, severe multi-year locust upsurges causing complete food system collapse may drive displacement.
- If food insecurity becomes severe, register with local authorities and WFP for food assistance programmes.

Emergency Contacts & Reporting

- FAO Desert Locust Information Service — global locust monitoring at locust.fao.org
- National Ministry of Agriculture — pest control coordination and relief programmes
- WFP — food assistance in severe food insecurity crises

CRITICAL WARNING: Never use home pesticides against swarms

Household or agricultural pesticides applied against a locust swarm are completely ineffective at swarm scale and create serious health hazards for the applicator, family, and livestock. Authorised control operations use specialised formulations at precise dosages applied by trained operators. Contact your agricultural authority immediately — do not attempt independent pesticide application.

SURVIVAL TIP: Hermetic grain storage protects your food supply

A hermetically sealed grain bag creates an oxygen-depleted environment that kills all insect life stages within weeks. Hermetically stored grain survives locust events, post-harvest pest infestations, and moisture damage. Initial investment in hermetic storage typically pays for itself within one season through reduced post-harvest losses.

Statistical Profile — Insect Infestation

The 2019-2021 Desert Locust crisis (the worst in 70 years) affected 23 countries across Africa, the Middle East, and South Asia, threatening food security for 42 million people. FAO coordinated a \$200M international response. A single desert locust swarm covering 1 km² can destroy crops that would feed 35,000 people for one day.

DISASTER TYPE 11

Landslide

Mass movement of earth, rock, and debris — triggered by rainfall, earthquakes, and deforestation

Quick Facts & Classification

Onset	Rapid to sudden — debris flows can travel at 20-30 m/s; slow creep landslides move centimetres per year
Trigger factors	Heavy or prolonged rainfall, seismic activity, volcanic activity, deforestation, slope cutting, saturation
Types	Rockfall, debris flow, mudslide, slump, translational slide, rotational slide
Primary hazards	Burial, trauma, asphyxiation, river damming and subsequent flood
Secondary hazards	Landslide dam outburst floods, road and infrastructure destruction, gas and power line rupture
Global median deaths	27 per event; individual events can kill hundreds
Key terms	landslide mudslide debris flow rockfall slope failure mass movement slope collapse

Before: Preparedness Actions

These actions should be completed before a disaster event occurs. Preparedness is the most effective way to reduce injury, death, and economic loss.

- Know whether your home is on or downslope of landslide-susceptible terrain. Look for warning signs: cracked walls, tilting trees and poles, unusual sounds, and water seeping from slopes.
- Contact your local geology or civil engineering authority for slope stability assessments in high-risk areas.
- Maintain or replant vegetation on slopes above your home — root systems dramatically improve slope stability.
- Avoid building on, near, or at the base of steep slopes, near drainage channels, or on natural landslide deposits.
- Install French drains or surface drainage to divert water away from steep slopes adjacent to your property.
- Know your evacuation route to safe ground. Ensure it does not cross drainages or the base of susceptible slopes.
- Keep a Go-Bag with essential items, medications, and documents.

During: Immediate Response Actions

These are the actions to take while the disaster event is actively occurring. Prioritise personal safety above all else.

- If you receive a landslide warning or observe warning signs (rumbling sounds, cracking of ground, flowing water carrying increasing sediment): evacuate to higher ground immediately.
- If landslide is imminent and escape is impossible: curl into a tight ball, protect your head with your arms, and move to the strongest part of the building (typically an interior corner).
- If caught in a debris flow: try to escape perpendicular to the flow direction — run to the side, not downhill.
- Stay out of valleys, drainage channels, and low-lying areas during and immediately after intense rainfall, especially in landslide-prone areas.
- Do not attempt to cross a debris-covered road or bridge until it has been inspected by engineers.
- Listen for changes in sounds: a roaring or rumbling that increases in volume typically indicates an approaching debris flow.

After: Recovery Actions

These actions apply in the hours, days, and weeks following the disaster event. Recovery involves physical safety, health, and psychological wellbeing.

- Stay away from the landslide area — additional slides are likely, particularly in the 24-72 hours following the initial event.
- Check for injured and trapped persons. Do not enter unstable areas to rescue without proper equipment and training.
- Report the landslide to local emergency management. Provide location, size, and whether persons are trapped.
- Check for and report broken gas, electrical, and water lines.
- Do not attempt to re-enter buildings that were in the path of or adjacent to the landslide until structurally assessed.
- Landslide dams can form rapidly in river channels — if a valley river suddenly stops flowing, a landslide dam may have formed upstream. Evacuate immediately — outburst flooding can follow with little warning.

Medical Guidance

- Crush syndrome: persons buried under debris may develop life-threatening kidney failure when released. Alert medical teams — do not rapidly extract buried persons without medical standby.
- Asphyxiation: debris can rapidly compress the chest. Even persons with their head above debris may be unable to breathe. Rapid, careful excavation of the chest is the priority.
- Hypothermia: wet, buried conditions rapidly reduce core body temperature. Insulate and warm the victim once excavated.
- Traumatic injuries from debris impact are common and may be masked by shock. All landslide survivors should receive medical evaluation.

Evacuation Procedures

- Evacuate immediately upon landslide warning — do not attempt to collect belongings.

- Follow routes that go away from slopes, not along valley floors.
- Establish a family meeting point on high, stable ground away from the affected slope system.

Emergency Contacts & Reporting

- National geological survey — landslide hazard maps and warnings
- Local civil defence — evacuation orders and rescue coordination
- Emergency services — rescue of trapped persons

CRITICAL WARNING: A landslide dam can burst without warning

When a landslide blocks a river, a dam forms. This dam can hold back weeks of water — and fail suddenly, releasing a catastrophic flood wave. If a river in a landslide-affected area stops flowing or reverses, immediately evacuate all areas downstream of the blockage. Do not wait for official confirmation.

SURVIVAL TIP: Cracks in your walls or ground may be your earliest warning

Slow-moving landslides begin long before the catastrophic phase. New cracks in walls, tilting of trees and poles, doors and windows that no longer close properly, and ground that appears to have shifted are early warning signs of slope instability. Report these observations to your local civil engineering authority immediately.

Statistical Profile — Landslide

Global disaster records show 776 landslide events (1900–2021). Median deaths: 27 — among the highest. Landslides are triggered by many other disaster types (earthquake, volcanic, excessive rainfall) and are classified as secondary hazards in many events. Deforestation dramatically increases landslide risk — studies show up to 20x higher slide frequency in deforested vs forested slopes.

DISASTER TYPE 12

Mass Movement (Dry)

Rock avalanches, dry rockfalls, and sand/dust storms — gravity-driven dry mass movements

Quick Facts & Classification

Types	Rockfall, rock avalanche, dry sand flow, loess collapse, avalanche (snow/ice)
Onset	Sudden — rockfalls and avalanches occur without warning
Trigger factors	Freeze-thaw cycles, thermal expansion, seismic activity, undercutting, weathering
Key regions	Mountain regions, arid rock-face terrain, permafrost areas (increasingly active due to warming)
Primary hazards	Impact trauma, burial, road and structure destruction
Key terms	rockfall rock avalanche dry mass movement avalanche debris collapse talus scree

Before: Preparedness Actions

These actions should be completed before a disaster event occurs. Preparedness is the most effective way to reduce injury, death, and economic loss.

- If living near cliff faces, steep rock outcrops, or mountain slopes: identify and stay clear of the runout zone — the area where fallen material would naturally travel.
- Install rockfall protection nets or barriers on high-risk slopes, working with your local civil engineering authority.
- For mountain travel: study terrain maps and identify avalanche paths and rockfall zones before entering mountain areas.
- Learn to read terrain for rockfall hazard: fresh angular debris at the base of a cliff, exposed fractured rock faces, and cliff overhangs are indicators of active rockfall.
- Avoid camping, parking, or resting at the base of steep rock faces, under cliffs, or in couloirs (steep mountain channels).
- Wear appropriate helmets when working or recreating in rockfall terrain.
- For avalanche terrain: carry a transceiver (beacon), probe, and shovel. Travel with trained companions. Take an avalanche safety course before entering avalanche terrain.

During: Immediate Response Actions

These are the actions to take while the disaster event is actively occurring. Prioritise personal safety above all else.

- If you observe or hear a rockfall or rock avalanche: move perpendicular to the slope, away from the path of falling material. Do not run downhill.

- Protect your head with your pack or arms. Seek shelter behind large stable boulders or a road cut.
- If caught in an avalanche: attempt to escape to the side. If engulfed: fight to stay near the surface, use swimming motions, protect a breathing space near your face, and conserve energy once movement stops — the snow sets like concrete.
- After movement stops in an avalanche: determine which way is up (saliva falls with gravity). Create a breathing space and expand the chest on inhalation before snow compacts. Call for help.
- During a dust or sand storm: seek shelter immediately in a building or vehicle. Cover your nose and mouth with a wet cloth. Turn off HVAC systems that draw outside air.

After: Recovery Actions

These actions apply in the hours, days, and weeks following the disaster event. Recovery involves physical safety, health, and psychological wellbeing.

- Do not re-enter the hazard zone until it has been inspected by geotechnical engineers.
- Report all rockfalls and mass movements to civil engineering authorities — even small events may indicate larger instability.
- After an avalanche: rapidly organised rescue within the first 15 minutes gives buried victims a 90% chance of survival. This drops to 30% by 30 minutes. Immediate search by companions is critical.
- Dust storm aftermath: clean HVAC filters. Ventilate buildings thoroughly. Check and clean water and food storage.

Medical Guidance

- Avalanche burial: survival time in snow is approximately 90 minutes for a victim with an air pocket. Rescuers should work rapidly but systematically.
- Impact injuries from rockfall include severe traumatic brain injury and spinal injury. Stabilise the spine before movement. Call for medical evacuation.
- Dust inhalation: prolonged exposure to fine particulate matter in dust storms causes acute respiratory distress and may trigger or worsen asthma. Those with respiratory conditions should shelter indoors with windows sealed.

Evacuation Procedures

- Evacuate when warned of imminent rockfall or mass movement hazard.
- In mountain environments with fresh snow loading or rapid temperature changes (freeze-thaw), exercise heightened caution and consider pre-emptive descent.

Emergency Contacts & Reporting

- National geological survey — rockfall and mass movement monitoring
- Mountain rescue service — for casualties in mountain terrain
- Local civil engineering authority — hazard assessment and protection measures

CRITICAL WARNING: The runout zone extends much further than expected

Rock and snow avalanches travel at extreme velocities (100-200+ km/h for large events) and travel far beyond the base of the slope. The runout zone of a major rock avalanche can extend kilometres onto flat ground. Do not assume you are safe merely because you are away from the foot of the slope.

SURVIVAL TIP: Companion rescue saves avalanche victims — not rescue teams

The statistical chance of avalanche victim survival is determined primarily by whether companions begin searching immediately with transceivers. Professional rescue services typically arrive after the survival window has closed. Every person entering avalanche terrain must carry a transceiver, know how to use it, and practice beacon searches regularly.

Statistical Profile — Dry Mass Movement

Dry mass movements are among the most lethal per-event disaster types due to their suddenness and the difficulty of providing warning. Permafrost thaw driven by climate change is increasing rockfall frequency in mountain regions globally. The 1970 Huascarán rockslide in Peru (66,000 deaths) remains the deadliest mass movement event in recorded history.

DISASTER TYPE 13

Storm

Tropical cyclones, hurricanes, typhoons, severe thunderstorms, and tornadoes

Quick Facts & Classification

Types	Tropical cyclone/hurricane/typhoon, extratropical cyclone, severe thunderstorm, tornado, hailstorm, blizzard, sandstorm
Onset	Tropical cyclones: days of warning available; Tornadoes: minutes of warning; Thunderstorms: hours
Duration	Tornadoes: minutes; Thunderstorms: hours; Hurricanes: days; Blizzards: days
Wind speed thresholds	Tropical Storm: 63-118 km/h; Category 1 Hurricane: 119-153 km/h; Category 5: 252+ km/h
Primary hazards	Storm surge, high winds, flooding, tornadoes, lightning, hail
Secondary hazards	Power outages, coastal erosion, infrastructure destruction, freshwater contamination
Global median deaths	14 per event; major events kill thousands
Key terms	hurricane cyclone typhoon storm tornado thunderstorm wind storm surge blizzard hail lightning

Before: Preparedness Actions

These actions should be completed before a disaster event occurs. Preparedness is the most effective way to reduce injury, death, and economic loss.

- Know your location's storm risk: coastal zone (storm surge), tornado alley (tornadoes), or inland (wind/flood). Different risks require different preparations.
- Reinforce your home for high winds: inspect and upgrade roof anchoring, install storm shutters or board windows, and brace garage doors.
- TROPICAL CYCLONE: prepare a 72-hour emergency kit. Know your evacuation zone (A-F or 1-5) and your route to an inland shelter.
- TORNADO: identify the safest location in your home — lowest floor, interior room, away from windows. A basement is ideal. Pre-designate this location with all household members.
- LIGHTNING: install surge protectors for all electronics. Know the '30-30 rule': if the time between lightning flash and thunder is less than 30 seconds, seek shelter; wait 30 minutes after the last thunder before going outside.
- Clear your yard of loose items that can become projectiles in high winds: garden furniture, tools, pot plants, and decorative items.
- Charge all devices and portable battery banks before a storm.
- Know how to shut off utilities at the mains. Have manual tools available for after a storm when power tools may not be usable.

- Obtain storm insurance and document all property for claims.
- For coastal communities: know the storm surge risk. Storm surge, not wind, kills the majority of hurricane victims.

During: Immediate Response Actions

These are the actions to take while the disaster event is actively occurring. Prioritise personal safety above all else.

- **TROPICAL CYCLONE:** shelter in a pre-identified interior room on the lowest floor if not evacuating. Stay away from windows. Do not go outside during the eye of the storm — conditions will deteriorate again rapidly.
- **TORNADO WARNING:** seek shelter immediately in the lowest interior room or basement. Protect head and neck with arms. Do not seek shelter under an overpass — wind speeds are higher there.
- **TORNADO** — if in a vehicle: abandon the vehicle and seek shelter in a solid building. If no building is available: lie flat in a ditch or low-lying area away from trees and vehicles, covering head and neck.
- **LIGHTNING:** indoors is safest. Stay away from tall isolated trees, metal fences, open water, and hilltops. If caught outdoors: crouch low, stay away from tall objects, spread out (5m between group members), and remove metal backpacks.
- **STORM SURGE:** evacuate all surge zones as directed. Storm surge of 3-6 metres can overtop sea walls and travel kilometres inland. It is the deadliest component of a hurricane.
- Stay informed through official emergency broadcast radio. Do not rely solely on phone networks, which may be overloaded or fail.
- Do not attempt to drive through flooded or debris-covered roads.

After: Recovery Actions

These actions apply in the hours, days, and weeks following the disaster event. Recovery involves physical safety, health, and psychological wellbeing.

- Do not go outside until officially declared safe. Post-storm debris, downed power lines, and structural hazards are immediate threats.
- Treat all downed power lines as live. Keep a 10-metre exclusion zone from all downed wires and report immediately to the power utility.
- Document all damage with photographs before cleanup, for insurance and government assistance claims.
- Do not use portable generators, charcoal grills, or camp stoves indoors — carbon monoxide poisoning kills many people in the days after major storms.
- Be aware of mould growth in flood-affected structures — remove wet materials within 24-48 hours.
- Conduct a welfare check on all vulnerable neighbours immediately after the storm.
- Use text messages for communication — voice networks are typically congested after major storms.
- Do not consume tap water until the supply is officially declared safe.

Medical Guidance

- Carbon monoxide poisoning from indoor generator use is the most preventable post-storm cause of death. Symptoms: headache, dizziness, confusion, nausea. Evacuate immediately and call emergency services if suspected.
- Chainsaw and debris-clearing injuries: post-storm cleanup is one of the most dangerous periods. Use proper PPE (gloves, eye protection, steel-capped boots), never operate chainsaws alone, and rest regularly.
- Storm surge drowning: the primary cause of hurricane mortality. Treat all water immersion casualties as potential near-drowning and hypothermia cases requiring medical evaluation.
- Lightning strike: all lightning strike victims require urgent medical evaluation even if they appear uninjured. Secondary cardiac effects can develop hours after the strike.

Evacuation Procedures

- Tropical cyclone evacuation orders are mandatory in many jurisdictions. Comply immediately and early — late evacuation can be impossible once conditions deteriorate.
- Know your evacuation zone designation and the shelter locations for your zone before storm season.
- If you choose to shelter in place (for Category 1-2 hurricanes), ensure your structure is rated for the expected wind speed and you are not in a storm surge zone.
- Evacuate before storm conditions arrive. Evacuating through tropical cyclone conditions is extremely dangerous.

Emergency Contacts & Reporting

- National Meteorological Service — official storm warnings and tracks
- National Emergency Management Agency — evacuation orders and shelter locations
- Red Cross — storm shelter, food, and family reunification services
- Power utility emergency line — downed power lines and outage reporting

CRITICAL WARNING: Storm surge is the deadliest hurricane hazard — not the wind

In major hurricanes, storm surge — ocean water pushed inland by the storm's rotation — can reach 6 metres or more and penetrate many kilometres inland. Even a Category 1 hurricane can produce a deadly storm surge. Wind speed ratings do not reflect storm surge risk. Evacuate all coastal storm surge zones as directed, regardless of wind category.

SURVIVAL TIP: Your safest place in a tornado is underground

A basement or below-grade safe room provides the best tornado protection. Interior rooms on the lowest floor are the next best option. The absolute worst locations are mobile homes (always evacuate before a tornado threat), areas under bridges and overpasses (creates a wind tunnel effect), and vehicles. Drill your plan before storm season.

Statistical Profile — Storm

Storms are the second most frequent natural disaster : 4,496 events (1900–2021). Median deaths: 14 per event. Median damage: \$76M USD — the second highest after extreme temperature. Storm activity and intensity is increasing with ocean warming. The 1970 Bhola Cyclone (Bangladesh, 500,000 deaths) remains the deadliest storm on record.

DISASTER TYPE 14

Volcanic Activity

Eruptions, pyroclastic flows, ash fall, and lahars — geological hazards near active volcanoes

Quick Facts & Classification

Onset	Variable — from sudden phreatic explosions (no warning) to months of precursor activity
Hazard types	Lava flow, pyroclastic flow/surge, ash fall, lahar (volcanic mudflow), volcanic gas, ballistic projectiles, tsunami (flank collapse)
Zones	Exclusion zone (life-threatening), restricted zone (high risk), caution zone (ash and gas hazard)
Key regions	Pacific Ring of Fire, East African Rift, Iceland, Caribbean arc
Global median deaths	40 per event — second highest after drought
Most dangerous hazard	Pyroclastic flow — superheated gas and debris at 700°C travelling at 700+ km/h
Key terms	volcano eruption lava pyroclastic ash fall lahar volcanic gas sulfur dioxide volcanic alert

Before: Preparedness Actions

These actions should be completed before a disaster event occurs. Preparedness is the most effective way to reduce injury, death, and economic loss.

- Know the volcanic alert level system used by your national volcanological authority (typically: Normal/Green, Advisory/Yellow, Watch/Orange, Warning/Red).
- Know the hazard zones for all volcanoes near your community — maps are available from national geological surveys.
- Prepare a Go-Bag, including N95 masks (for ash fall) and goggles.
- Keep vehicles fuelled during periods of elevated volcanic alert — ash fall can make roads impassable within hours.
- Secure your water supply: volcanic ash contaminates open water sources rapidly. Store water in sealed containers.
- Know the lahar risk for your area: lahars (volcanic mudflows) travel along river valleys and can occur during or long after an eruption when rain mobilises volcanic deposits. Do not assume a lahar risk has passed because the eruption has ended.
- Understand eruption alert level thresholds and what actions are required at each level. Know your community's designated evacuation assembly points.

During: Immediate Response Actions

These are the actions to take while the disaster event is actively occurring. Prioritise personal safety above all else.

- Follow all official evacuation orders immediately. Volcanic hazard zones are determined by expert volcanological analysis — do not assume the exclusion zone is excessive.
- ASH FALL: remain indoors with windows and doors closed. Wear an N95 mask if you must go outside. Cover nose and mouth with a damp cloth if a mask is unavailable. Protect eyes with goggles.
- ASH FALL: clear ash from rooftops regularly — wet ash is extremely heavy (up to 1,000 kg/m²) and can cause structural roof collapse. Do not allow ash to accumulate.
- LAHAR WARNING: move immediately to high ground — lahars travel at 20-40 km/h along valley floors and are virtually impossible to outrun if caught on the valley floor.
- LAVA FLOW: evacuate as directed. Lava moves slowly enough to evacuate from, but it destroys everything in its path absolutely and irreversibly.
- PYROCLASTIC FLOW: if in the path of a pyroclastic flow, there is no survivable shelter in its direct path. The only survival strategy is to not be in the exclusion zone when an eruption occurs.
- VOLCANIC GAS (sulfur dioxide, hydrogen sulfide): seek higher ground — volcanic gases are denser than air and accumulate in low-lying areas, particularly valleys and caves.
- Monitor your official alert radio continuously. Alert levels can change rapidly.

After: Recovery Actions

These actions apply in the hours, days, and weeks following the disaster event. Recovery involves physical safety, health, and psychological wellbeing.

- Ash fall cleanup: use N95 masks and goggles. Do not use water to clean ash from rooftops (it becomes concrete-like when wet). Use dry sweeping.
- Do not consume food or water that has been exposed to volcanic ash until tested or treated.
- Follow official guidance on water safety — volcanic ash contaminates surface water supplies and can block municipal water infrastructure.
- Lahar risk persists for years after an eruption. Heavy rainfall can mobilise volcanic deposits long after the eruption has ceased. Maintain ongoing awareness of lahar warning systems.
- Structures in ash-fall areas must be assessed for structural integrity before re-entry — ash loading can cause roof collapse.
- Monitor respiratory health in the weeks following ash fall — volcanic ash causes serious respiratory conditions including silicosis with repeated exposure.

Medical Guidance

- Ash inhalation: volcanic ash particles are sharp, chemically reactive, and can penetrate deep into the lungs. Prolonged exposure without respiratory protection causes silicosis. Anyone with significant ash inhalation should receive medical evaluation.
- Eye injury: volcanic ash particles cause corneal abrasion. Flush eyes immediately with clean water. Use goggles in ash-fall conditions.
- Burns: lava and pyroclastic deposits remain thermally hazardous for extended periods. Do not approach recently deposited volcanic material without specialist equipment.

- Volcanic gas poisoning (H₂S): hydrogen sulfide smells of rotten eggs at low concentrations but causes olfactory fatigue at higher concentrations — you stop smelling it before it kills you. Evacuate any area where H₂S is suspected without relying on your sense of smell.

Evacuation Procedures

- Evacuate when ordered — do not wait to observe the eruption. Many volcanic eruption fatalities occur among those who delayed evacuation.
- Follow designated routes. Avoid river valleys in lahar risk areas.
- Do not attempt to observe the eruption from 'safe' vantage points near the exclusion zone — eruption intensification can be sudden.
- Carry your N95 masks, goggles, and Go-Bag. Evacuation may be prolonged — months to years in severe cases.

Emergency Contacts & Reporting

- National volcanological observatory — official eruption alerts and exclusion zone status
- Civil defence and emergency management — evacuation coordination
- National health authority — ash health guidance and medical support

CRITICAL WARNING: There is no shelter from a direct pyroclastic flow

A pyroclastic flow is a superheated mixture of gas, ash, and volcanic rock at up to 700 degrees Celsius, travelling at up to 700 km/h. No building, vehicle, or structure can withstand a direct pyroclastic flow. The 1902 eruption of Mount Pelee killed 30,000 people in minutes. The only survival strategy is to be outside the exclusion zone before the eruption.

SURVIVAL TIP: Clear ash from your roof — wet ash weighs as much as concrete

Volcanic ash accumulating on a roof to a depth of just 10-15 cm can weigh as much as 100-150 kg per square metre. After heavy ash fall, many building roofs collapse under this load, particularly flat or shallow-pitched roofs. Clear ash regularly during an ash fall event using a broom. Do not add water — wet ash is far heavier.

Statistical Profile — Volcanic Activity

Global disaster records show 265 volcanic events (1900–2021). Median deaths: 40 per event — the second highest of all disaster types. The 1815 Tambora eruption (Indonesia, 71,000+ deaths and subsequent global climate impact) remains the most impactful volcanic event in recorded history. Modern volcano monitoring has dramatically improved evacuation lead times.

DISASTER TYPE 15

Wildfire

Uncontrolled vegetation fires — increasingly frequent and severe due to climate change

Quick Facts & Classification

Onset	Rapid — a wildfire can travel faster than a person can run (6+ km/h in light fuel; up to 23 km/h with wind)
Primary drivers	Drought, high temperatures, low humidity, wind, accumulated dry fuel (vegetation), ignition source
Types	Crown fire (treetop), surface fire, ground fire (peat), interface fire (wildland-urban)
Primary hazards	Direct flame, radiant heat, smoke inhalation, ember spotting (fires ignited by airborne embers ahead of the fire front)
Secondary hazards	Post-fire landslide (destabilised slopes), flooding (destroyed vegetation), air quality crisis
Global median damage	\$102M USD — third highest of all disaster types
Global median deaths	6 per event (but major events kill hundreds)
Key terms	wildfire bushfire forest fire vegetation fire smoke ember spot fire evacuation

Before: Preparedness Actions

These actions should be completed before a disaster event occurs. Preparedness is the most effective way to reduce injury, death, and economic loss.

- Create a DEFENSIBLE SPACE around your home: remove dry vegetation within 10m of your structure; thin and trim vegetation between 10-30m; clear roofs, gutters, and decks of leaves and debris.
- Use fire-resistant building materials on roofs, walls, and decks where possible. Block all ember entry points: gaps in eaves, roof vents, and under doors.
- Store firewood, propane tanks, and flammable materials away from your home structure (minimum 10m).
- Keep your gutters clean of dry leaves — burning embers landing in gutters filled with dry material ignite homes before the fire front arrives.
- Prepare your home for rapid departure: store your Go-Bag in your vehicle, park facing the exit direction, and keep your fuel tank at least half full during fire season.
- Know your community's fire emergency plan: evacuation routes (multiple, in case one is blocked by fire), designated assembly points, and shelter locations.
- Identify your local warning system: emergency alerts, sirens, and how local authorities issue evacuation orders.
- Monitor fire danger ratings daily during fire season. Know what a 'Catastrophic' or 'Code Red' rating means for your region.

- Ensure all household members know the plan and can execute it independently, including children.
- Obtain home and contents insurance that includes wildfire coverage. Photograph all contents.

During: Immediate Response Actions

These are the actions to take while the disaster event is actively occurring. Prioritise personal safety above all else.

- Evacuate as early as possible when a warning is issued. Late evacuation in the path of a rapidly advancing fire is fatal.
- If ordered to evacuate: go immediately. Do not wait to collect additional belongings. Your life is not worth any object.
- If you cannot evacuate (trapped): shelter in the most fire-resistant building available. Close all doors and windows. Fill sinks and baths with water. Remove all curtains and move furniture away from windows.
- Smoke is the primary immediate danger: close all windows and doors to reduce smoke infiltration. Seal gaps with wet towels.
- Avoid driving through smoke — smoke causes rapid and complete loss of visibility. If caught in dense smoke while driving, pull off, turn off the engine, close all vents, and stay in the vehicle.
- Do not shelter in your vehicle as a last resort in the path of a fire front — a vehicle's fuel tank, tyres, and plastic components can ignite. Seek a structure or cleared area.
- If all other options are exhausted and fire overruns your location: lie face-down in a depression or ditch on cleared ground, covering your body with non-synthetic clothing or soil.
- Monitor your official emergency radio or app continuously — fire conditions change rapidly and evacuation orders may be issued with very short notice.

After: Recovery Actions

These actions apply in the hours, days, and weeks following the disaster event. Recovery involves physical safety, health, and psychological wellbeing.

- Do not return home until officially declared safe — fire can reignite in smouldering ground or structures for days after the main front passes.
- Assume all tap water is unsafe until your water authority confirms otherwise — wildfires melt and contaminate plastic water pipes.
- Wear a P2/N95 mask when re-entering fire-affected areas — ash and soil from burned areas contain toxic compounds including heavy metals.
- Wear protective gloves, long sleeves, and closed-toe shoes during post-fire cleanup.
- Burned slopes are highly susceptible to landslide and flooding for years after a fire — the protective root structure of vegetation has been destroyed. Follow rainfall warnings diligently.
- Document all damage with photographs and video before any cleanup, for insurance claims and government assistance.
- Check in with your mental health: wildfire survivors experience high rates of PTSD, anxiety, and grief. Seek support.
- Check air quality index (AQI) before spending time outdoors in fire-affected regions — smoke can cause serious respiratory harm at levels well below visible haze.

Medical Guidance

- Smoke inhalation: carbon monoxide and hydrogen cyanide in wildfire smoke are rapidly lethal at high concentrations. Move casualties to fresh air immediately. Seek emergency medical care for all significant smoke exposure — delayed pulmonary oedema can develop hours later.
- Burns: do not apply ice, butter, or toothpaste to burns. Cool with clean cool (not cold) running water for 20 minutes. Cover with a clean non-stick dressing. Seek immediate medical care for all but the most minor burns.
- Eye irritation from ash and smoke: flush eyes with clean water. Avoid rubbing. Wear goggles when in smoke or ash conditions.
- Post-fire air quality illness: fine particulate matter (PM2.5) from wildfire smoke causes and exacerbates asthma, COPD, and cardiovascular disease. Those with pre-existing conditions should monitor AQI and shelter indoors during high-PM periods.

Evacuation Procedures

- Evacuate on WATCH level — do not wait for an ORDER. Roads in fire zones can be cut off by fire, accidents (due to mass evacuation traffic), or downed power lines.
- Know multiple exit routes from your community. Practice your primary and alternate routes before fire season.
- If your route is blocked by fire and you cannot turn back: do not drive through the fire. Seek the nearest structure or open cleared area.
- Pet and livestock evacuation requires advance planning — arrange transport and destination in advance. Do not delay your own evacuation for animals if you have no plan.

Emergency Contacts & Reporting

- National fire authority — fire danger ratings, incident updates, and evacuation orders
- Local emergency management — official evacuation orders and shelter locations
- Emergency broadcast radio — the most reliable source of information during power outages
- Red Cross — emergency shelter and family reunification

CRITICAL WARNING: Ember attack ignites homes — not the fire front

In most wildland-urban interface fires, homes are not ignited by the main fire front directly. They are ignited by burning embers (firebrands) that travel ahead of the fire on the wind, landing on roofs, in gutters, and in gaps in eaves. This is why defensible space and ember-proofing your home matters far more than the fire's distance from your property. A home 2km from a fire front can be ignited by embers in minutes.

SURVIVAL TIP: Leave early — never late

Every wildfire fatality analysis reaches the same conclusion: people who died trying to evacuate waited too long. A fire that is 20km away under dry, hot, windy conditions can reach your home in under an hour. When a watch is issued, load your car and be ready. When an order is issued, leave immediately without debate. The cost of an unnecessary evacuation is one night in a shelter. The cost of a late evacuation can be your life.

Statistical Profile — Wildfire

Global disaster records show 471 wildfire events (1900–2021). Median damage: \$102M USD — third highest of all disaster types. The 2019-2020 Australian Black Summer fires burned 18.6 million hectares, killed 34 people directly, and an estimated 417 people from smoke-related illness. California's 2018 Camp Fire (85 deaths, \$16.5B damage) illustrates the catastrophic potential of wildland-urban interface fires.